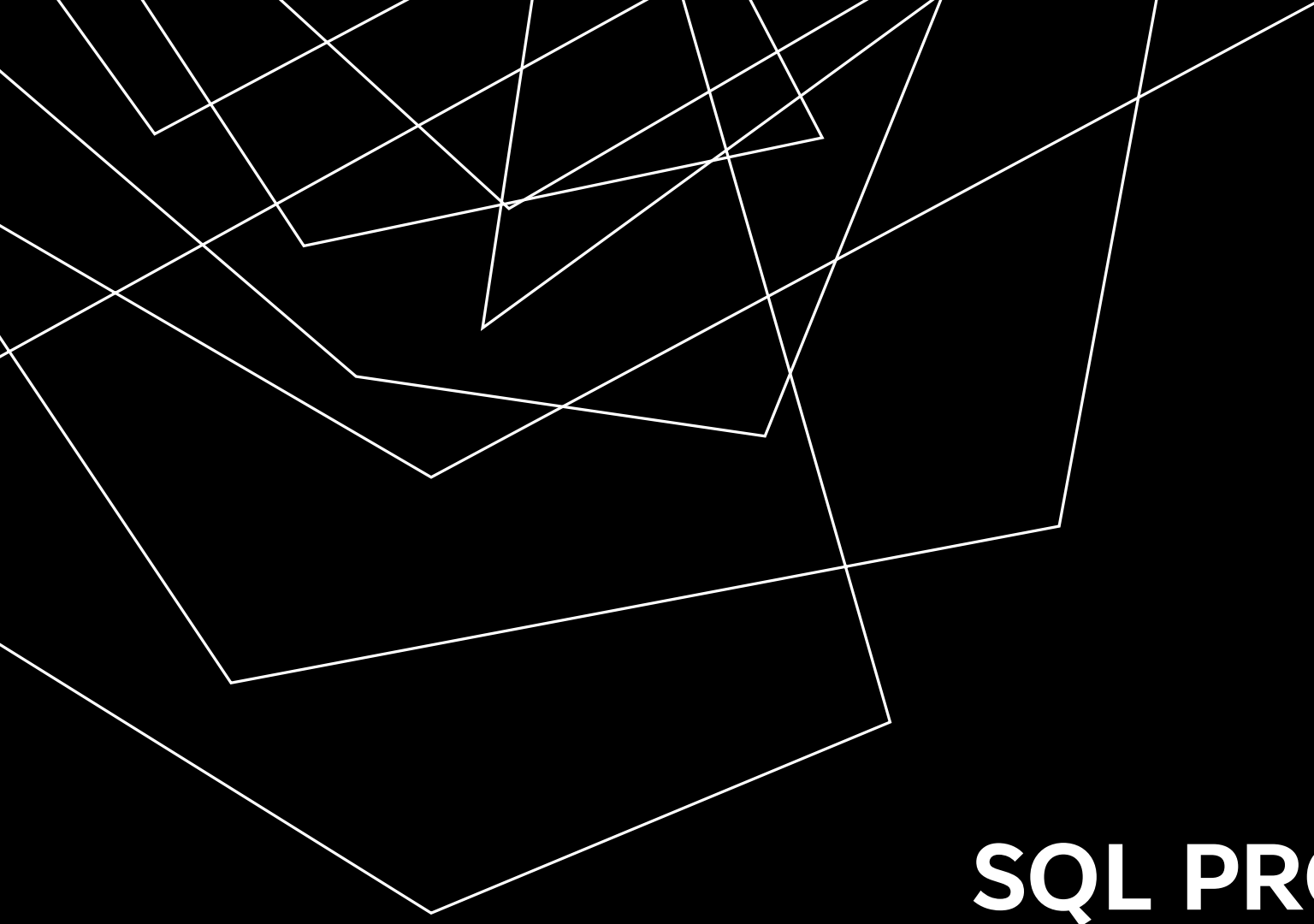


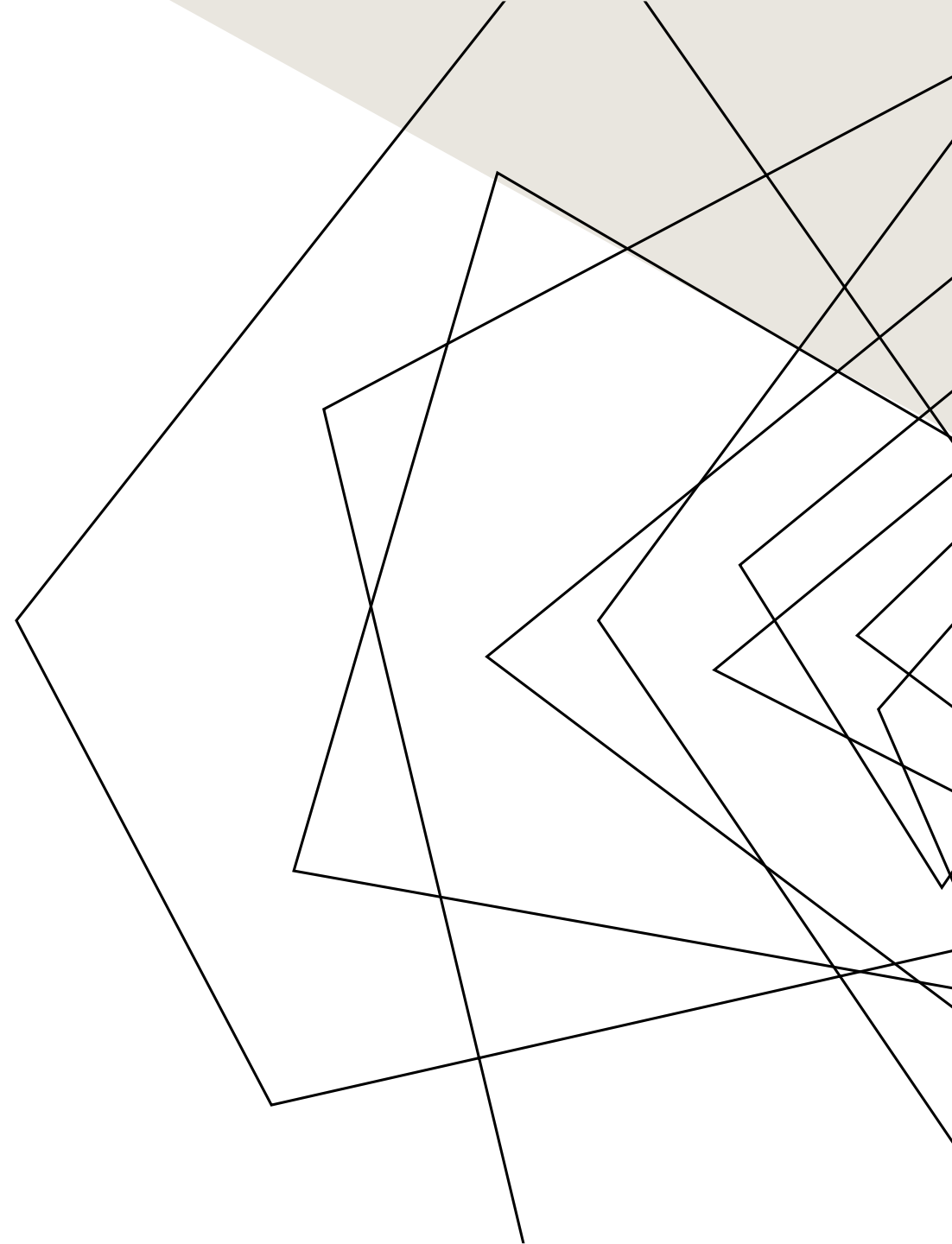
Abstract geometric lines in white on a black background, forming various polygons and intersecting lines.

SQL PROJECT

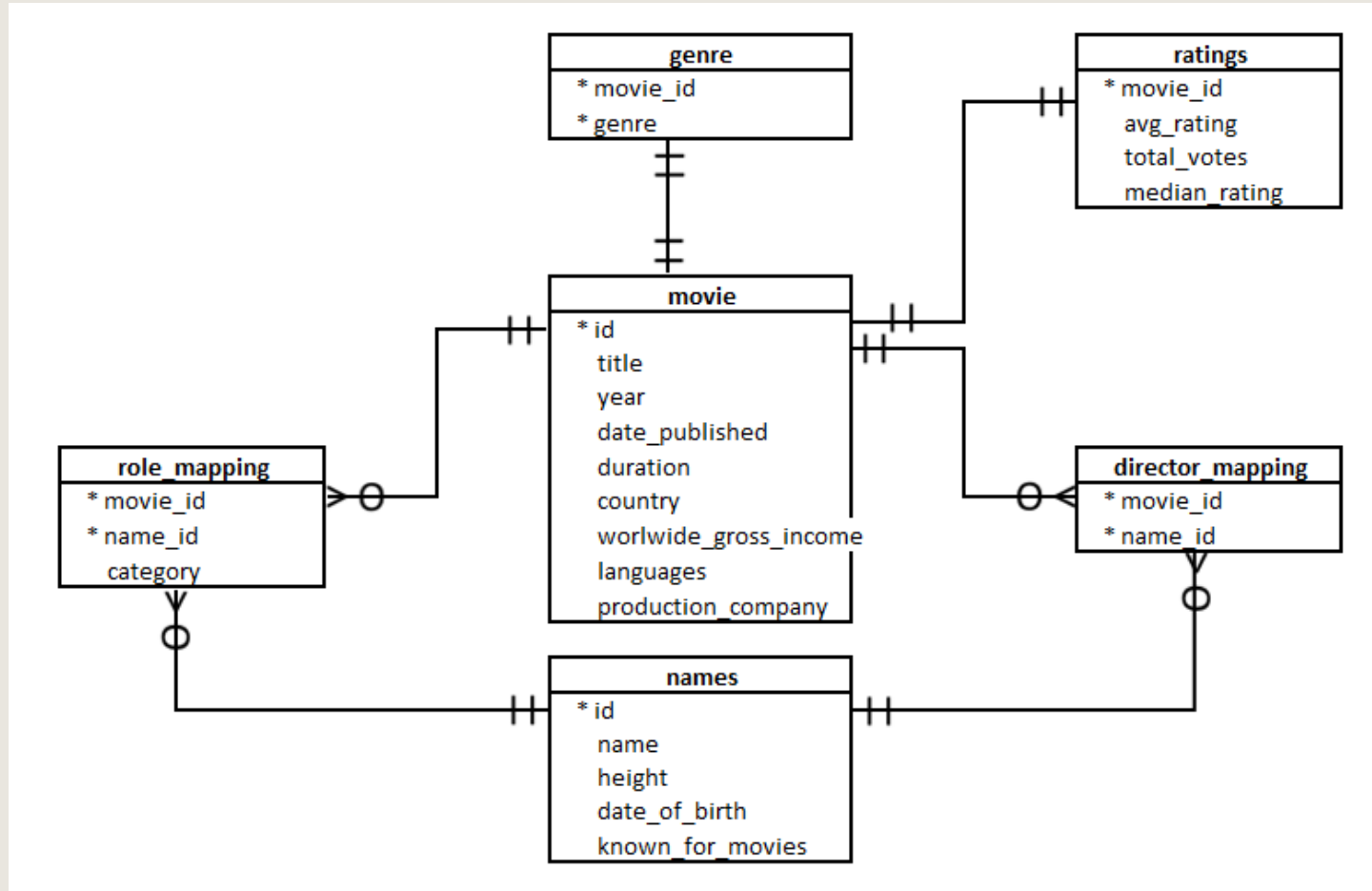
IMDB MOVIE DATABASE ANALYSIS

PROJECT OVERVIEW

The objective of the IMDb Movie Database Analysis Using SQL project is to solve a series of 29 questions that range from basic to advanced levels, all related to querying and analyzing the IMDb dataset. This project serves as a practical exercise to enhance my SQL skills by working with six interconnected tables: **`director mapping`**, **`genre`**, **`movie`**, **`names`**, **`rating`**, and **`role mapping`**. By addressing these questions, I aim to gain a deeper understanding of movie data, including aspects like genres, directors, ratings, and roles across various countries. This hands-on approach provides a comprehensive learning experience, allowing me to apply theoretical SQL knowledge to real-world data scenarios and improve my data analysis capabilities.



TABLES OVERVIEW



Q1- FIND THE TOTAL NUMBER OF ROWS IN EACH TABLE OF THE SCHEMA?

QUERY

```
select count(*) as movie from movie;
```

```
select count(*) as Director_mapping from director_mapping;
```

```
select count(*) as name from names;
```

```
select count(*) as rating from ratings;
```

OUTPUT

Result Grid	
	movie
▶	7997

3867

25735

7997

Q2- Find the total number of movies released each year? How does the trend look month wise?

QUERY

```
/* Output format for the first part:
+-----+-----+
| Year      | number_of_movies|
+-----+-----+
| 2017      | 2134             |
| 2018      | .                |
| 2019      | .                |
+-----+-----+*/

-- Type your code below:

select year, count(id) as number_of_movies from movie
group by year;
```

OUTPUT

	year	number_of_movies
▶	2017	3052
	2018	2944
	2019	2001

```
/* Output format for the second part of the question:
+-----+-----+
| month_num | number_of_movies|
+-----+-----+
| 1         | 134              |
| 2         | 231              |
| .         | .                |
+-----+-----+*/

-- Type your code below:

select month(date_published) as month_num, count(id) as number_of_movies from movie
group by month_num
order by month_num;
```

	month_num	number_of_movies
▶	1	804
	2	640
	3	824
	4	680
	5	625
	6	580
	7	493
	8	678
	9	809
	10	801
	11	625
	12	438

Q3- HOW MANY MOVIES WERE PRODUCED IN THE USA OR INDIA IN THE YEAR 2019?

QUERY

```
Select count(id) as number_of_movies from movie where year=2019 and (country='USA' or country='india');
```

OUTPUT

	number_of_movies
▶	887

Q4- Find the unique list of the genres present in the data set?

QUERY

```
select distinct genre from genre;
```

OUTPUT

	genre
►	Drama
	Fantasy
	Thriller
	Comedy
	Horror
	Family
	Romance
	Adventure
	Action
	Sci-Fi
	Crime
	Mystery
	Others

Q5-HOW MANY MOVIES BELONG TO ONLY ONE GENRE?

QUERY

```
select genre, count(movie_id) as count_of_movies from genre group by genre order by count_of_movies desc limit 5;
```

OUTPUT

	genre	count_of_movies
►	Drama	4285
	Comedy	2412
	Thriller	1484
	Action	1289
	Horror	1208

Q6- How many movies belong to only one genre?

QUERY

```
select count(*) as num_movie_single_genre
from(
  select movie_id
  from genre
  group by movie_id
  having count(*) =1
) as single_genre_movies;
```

OUTPUT

	num_movie_single_genre
▶	3289

Q7- WHAT IS THE AVERAGE DURATION OF MOVIES IN EACH GENRE?

QUERY

```
Select g.genre,  
       avg(m.duration) as avg_duration  
from  
       genre as g  
inner join movie as m  
on g.movie_id=m.id  
group by genre  
order by genre;
```

OUTPUT

	genre	avg_duration
▶	Action	112.8829
	Adventure	101.8714
	Comedy	102.6227
	Crime	107.0517
	Drama	106.7746
	Family	100.9669
	Fantasy	105.1404
	Horror	92.7243
	Mystery	101.8000
	Others	100.1600
	Romance	109.5342
	Sci-Fi	97.9413
	Thriller	101.5761

Q8- What is the rank of the 'thriller' genre of movies among all the genres in terms of number of movies produced?

QUERY

```
with summary as
(
  select genre,
         count(movie_id) as movie_count,
         Rank() over (order by count(movie_id) desc) as genre_rank
  from genre
  group by genre
)
select * from summary where genre ="thriller";
```

OUTPUT

	genre	movie_count	genre_rank
►	Thriller	1484	3

Q9-HOW MANY MOVIES BELONG TO ONLY ONE GENRE?

QUERY

```
select
MIN(avg_rating) as min_avg_rating,
MAX(avg_rating) as max_avg_rating,
MIN(median_rating) as min_median_rating,
MAX(median_rating) as max_median_rating,
MIN(total_votes) as min_total_votes,
MAX(total_votes) as max_total_votes
from ratings;
```

OUTPUT

	min_avg_rating	max_avg_rating	min_median_rating	max_median_rating	min_total_votes	max_total_votes
▶	1.0	10.0	1	10	100	725138

Q10- Find the total number of movies released each year? How does the trend look month wise?

QUERY

```
SELECT
  m.title,
  AVG(r.avg_rating) AS avg_rating,
  RANK() OVER (ORDER BY AVG(r.avg_rating) DESC) AS movie_rank
FROM
  ratings AS r
INNER JOIN
  movie AS m ON r.movie_id = m.id
GROUP BY
  m.title
ORDER BY
  avg_rating DESC
LIMIT 10;
```

OUTPUT

	title	avg_rating	movie_rank
▶	Kirket	10.00000	1
	Love in Kilnerry	10.00000	1
	Gini Helida Kathe	9.80000	3
	Runam	9.70000	4
	Fan	9.60000	5
	Android Kunjappan Version 5.25	9.60000	5
	Yeh Suhaagraat Impossible	9.50000	7
	Safe	9.50000	7
	The Brighton Miracle	9.50000	7
	Shibu	9.40000	10

Q11- SUMMARIZE THE RATINGS TABLE BASED ON THE MOVIE COUNTS BY MEDIAN RATINGS.

QUERY

```
select
  median_rating,
  count(movie_id) as movie_count
from ratings
group by median_rating
order by median_rating;
```

OUTPUT

	median_rating	movie_count
▶	1	94
	2	119
	3	283
	4	479
	5	985
	6	1975
	7	2257
	8	1030
	9	429
	10	346

Q12- Which production house has produced the most number of hit movies (average rating > 8)?

QUERY

```
SELECT
  m.production_company AS Production_company,
  count(m.id) AS movie_count,
  rank() over (order by avg(r.avg_rating) desc) as Prod_company_rank
FROM
  movie as m
INNER JOIN
  ratings AS r on m.id = r.movie_id
GROUP BY
  m.production_company
HAVING
  avg(r.avg_rating) > 8
ORDER BY
  prod_company_rank
limit 10;
```

OUTPUT

	Production_company	movie_count	Prod_company_rank
►	Archway Pictures	1	1
	A square productions	1	1
	Bestwin Production	1	3
	SLN Cinemas	1	4
	Epiphany Entertainments	1	5
	Eastpool Films	1	5
	Piecewing Productions	1	5
	Lovely World Entertainment	1	8
	Manikya Productions	1	8
	Kaargo Cinemas	1	8

Q13- HOW MANY MOVIES RELEASED IN EACH GENRE DURING MARCH 2017 IN THE USA HAD MORE THAN 1,000 VOTES?

QUERY

```
Select
    genre.genre,
    movie.country,
    count(movie.id)
from
    movie
Inner join
    genre
on
    genre.movie_id = movie.id
where
    movie.country = 'USA'
group by
    genre.genre, movie.country;
```

OUTPUT

	genre	country	count(movie.id)
►	Horror	USA	609
	Drama	USA	1025
	Comedy	USA	634
	Action	USA	315
	Sci-Fi	USA	163
	Thriller	USA	556
	Romance	USA	235
	Crime	USA	245
	Adventure	USA	182
	Mystery	USA	178
	Family	USA	84
	Fantasy	USA	103
	Others	USA	30

Q14- Find movies of each genre that start with the word 'The' and which have an average rating > 8?

QUERY

```
Select
  movie.title as title,
  genre.genre as genre,
  ratings.avg_rating as avg_rating,
  dense_rank() OVER (PARTITION BY genre ORDER BY avg_rating) AS rank_number
From movie
Inner join genre
on movie.id=genre.movie_id
Inner join ratings
on genre.movie_id=ratings.movie_id
where
  avg_rating > 8 and
  title like 'The%';
```

OUTPUT

	title	genre	avg_rating	rank_number
▶	Theeran Adhigaaram Ondru	Action	8.3	1
	Theeran Adhigaaram Ondru	Crime	8.3	1
	The Gambinos	Crime	8.4	2
	The Irishman	Crime	8.7	3
	The King and I	Drama	8.2	1
	The Gambinos	Drama	8.4	2
	The Mystery of Godliness: The Sequel	Drama	8.5	3
	The Irishman	Drama	8.7	4
	The Blue Elephant 2	Drama	8.8	5
	The Colour of Darkness	Drama	9.1	6
	The Brighton Miracle	Drama	9.5	7
	The Blue Elephant 2	Horror	8.8	1
	The Blue Elephant 2	Mystery	8.8	1
	The King and I	Roma...	8.2	1
	Theeran Adhigaaram Ondru	Thriller	8.3	1

Q15- THE MOVIES RELEASED BETWEEN 1 APRIL 2018 AND 1 APRIL 2019, HOW MANY WERE GIVEN A MEDIAN RATING OF 8?

QUERY

```
Select
  dense_rank() over (partition by title order by title) as row_num,
  movie.title as title,
  movie.date_published as Released_date,
  ratings.median_rating as median_rating
from
  movie
inner join
  ratings
on
  ratings.movie_id=movie.id
where
  movie.date_published BETWEEN '2018-04-01' AND '2019-04-01'
  AND ratings.median_rating > 8;
```

OUTPUT

	count_of_movies
▶	361

Q16- Do German movies get more votes than Italian movies?

QUERY

```
Select
    count(ratings.total_votes),
    movie.country
from
    movie
inner join
    ratings
on movie.id = ratings.movie_id
where movie.country in ('Germany', 'Italy')
group by
    movie.country;
```

OUTPUT

	count(ratings.total_votes)	country
▶	146	Germany
	123	Italy

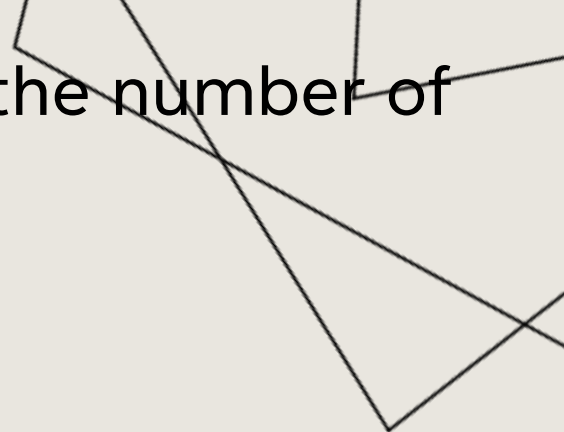
Q17-WHICH COLUMNS IN THE NAMES TABLE HAVE NULL VALUES?

QUERY

```
SELECT
    SUM(CASE WHEN name IS NULL THEN 1 ELSE 0 END) AS name_nulls,
    SUM(CASE WHEN height IS NULL THEN 1 ELSE 0 END) AS height_nulls,
    SUM(CASE WHEN date_of_birth IS NULL THEN 1 ELSE 0 END) AS date_of_birth_nulls,
    SUM(CASE WHEN known_for_movies IS NULL THEN 1 ELSE 0 END) AS known_for_movies_nulls
FROM
    names;
```

OUTPUT

	name_nulls	height_nulls	date_of_birth_nulls	known_for_movies_nulls
▶	0	17335	13431	15226



Q18- Which are the top three production houses based on the number of votes received by their movies?

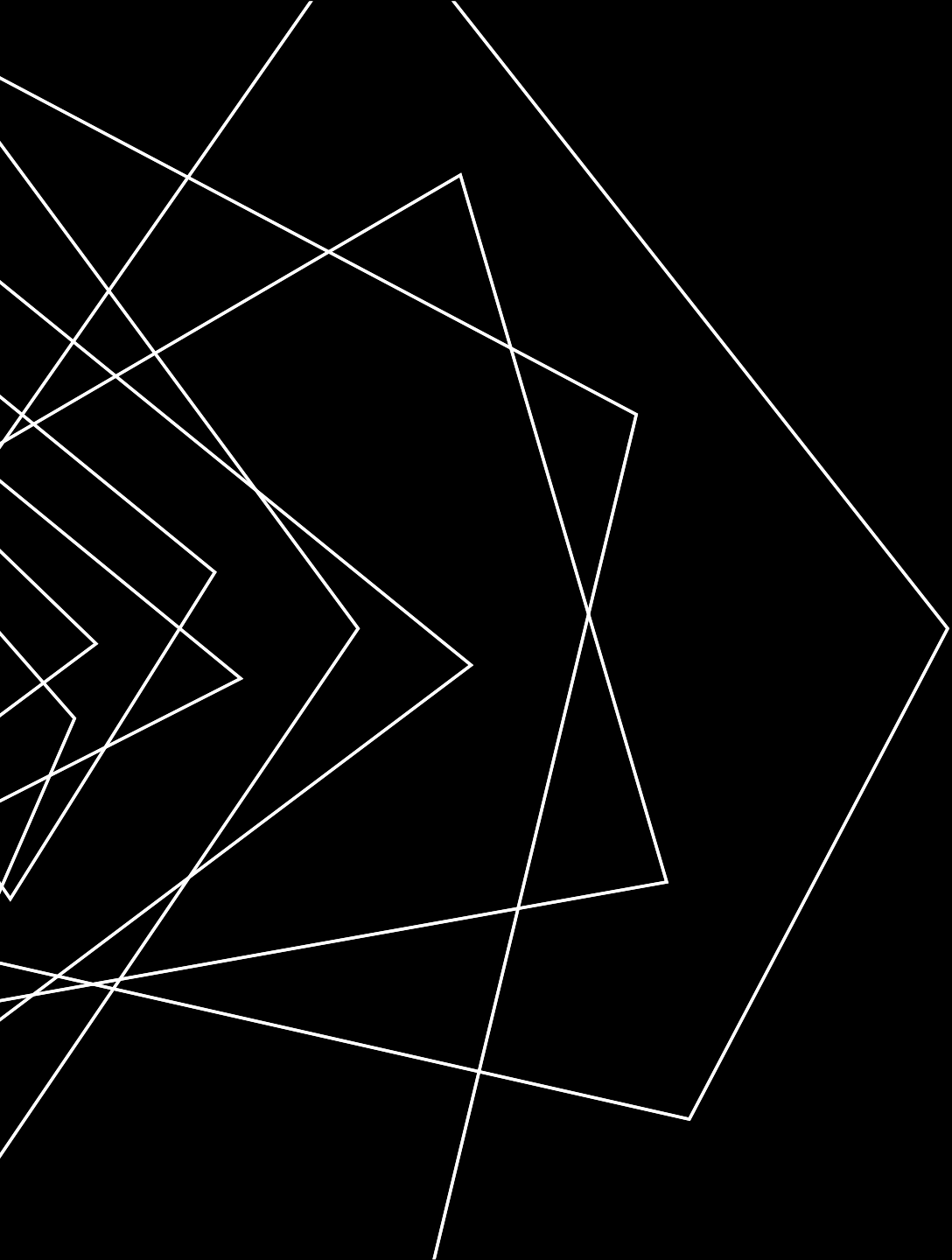
QUERY

```
Select
  movie.production_company as Production_company,
  ratings.total_votes as vote_count,
  rank () over (order by ratings.total_votes desc) as Prod_comp_rank
from
  movie
inner join
  ratings on movie.id = ratings.movie_id
limit 3;
```

OUTPUT

	Production_company	vote_count	Prod_comp_rank
▶	Marvel Studios	725138	1
	Marvel Studios	602792	2
	Twentieth Century Fox	586106	3





THANK YOU

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